

Active Reconnaissance

Active reconnaissance requires you to make some kind of contact with your target. This contact can be a phone call or a visit to the target company under some pretence to gather more information, usually as part of social engineering. Alternatively, it can be a direct connection to the target system, whether visiting their website or checking if their firewall has an SSH port open.

Web Browser

On the transport level, the browser connects to:

- TCP port 80 by default when the website is accessed over HTTP
- TCP port 443 by default when the website is accessed over HTTPS

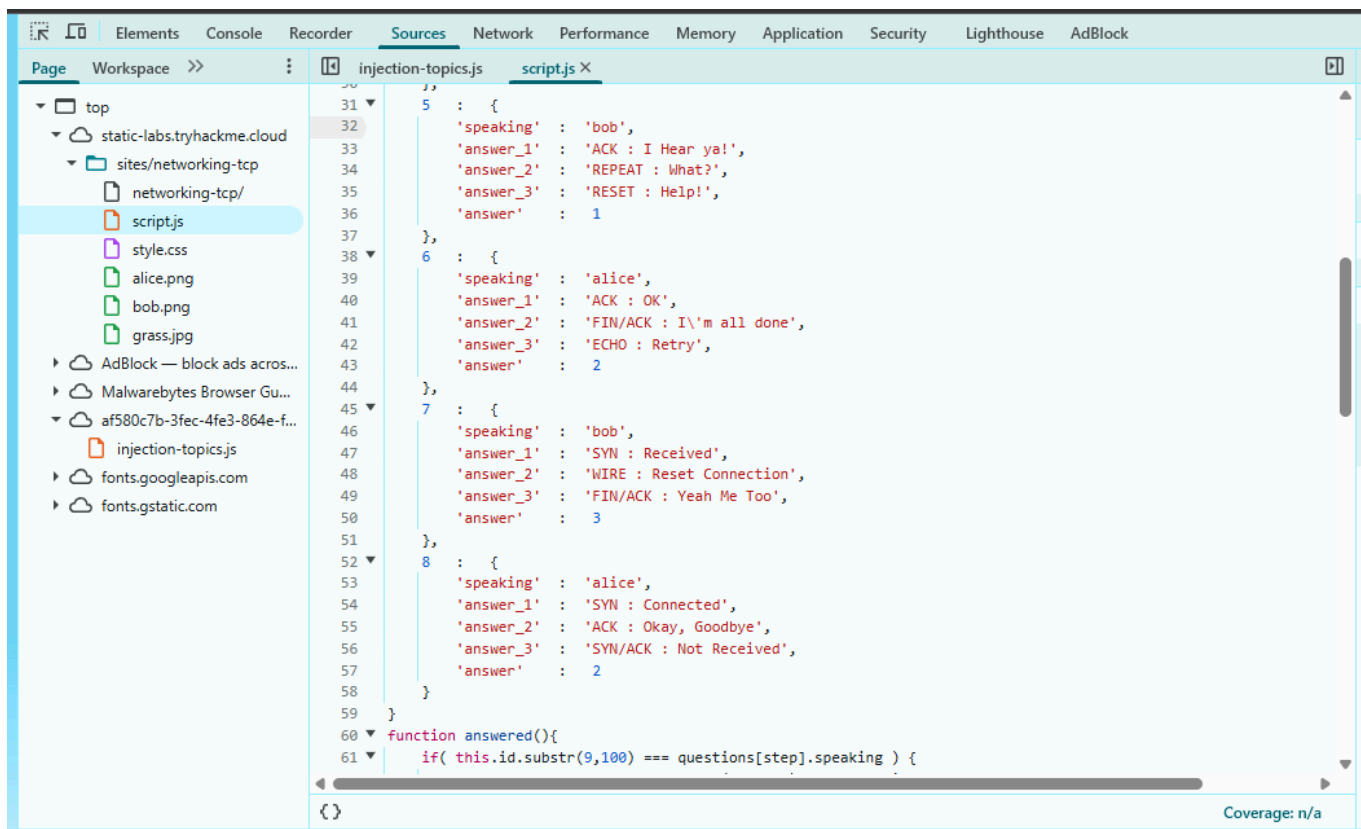
it is possible to use custom ports to access a service. For instance, <https://127.0.0.1:8834/> will connect to 127.0.0.1 (localhost) at port 8834 via HTTPS protocol.

Developer Tools lets you inspect many things that your browser has received and exchanged with the remote server. For instance, you can view and even modify the JavaScript (JS) files, inspect the cookies set on your system and discover the folder structure of the site content.

There are also plenty of add-ons for Firefox and Chrome that can help in penetration testing. Here are a few examples:

- **FoxyProxy** lets you quickly change the proxy server you are using to access the target website. This browser extension is convenient when you are using a tool such as Burp Suite or if you need to switch proxy servers regularly. You can get FoxyProxy for Firefox from [here\(opens in new tab\)](#).
- **User-Agent Switcher and Manager** gives you the ability to pretend to be accessing the webpage from a different operating system or different web browser. In other words, you can pretend to be browsing a site using an iPhone when in fact, you are accessing it from Mozilla Firefox. You can download User-Agent Switcher and Manager for Firefox [here\(opens in new tab\)](#).
- **Wappalyzer** provides insights about the technologies used on the visited websites. Such extension is handy, primarily when you collect all this information while browsing the website like any other user. A screenshot of Wappalyzer is shown below. You can find Wappalyzer for Firefox [here\(opens in new tab\)](#).

i used dev tools to view the Java script if a website to figure out how many questions are asked



Ping

`ping MACHINE_IP` or `ping HOSTNAME`

Hence, you might consider `ping -c 10 MACHINE_IP` if you just want to send ten packets. This is equivalent to `ping -n 10 MACHINE_IP` on a MS Windows system.

If a target responds to ping:

From a penetration testing point of view, we will try to discover more about this target. We will try to learn as much as possible, for example, which ports are open and which services are running.

when we don't get a ping reply back, there are a few explanations that would explain why we didn't get a ping reply, for example:

- The destination computer is not responsive; possibly still booting up or turned off, or the OS has crashed.
- It is unplugged from the network, or there is a faulty network device across the path.
- A firewall is configured to block such packets. The firewall might be a piece of software running on the system itself or a separate network appliance. Note that MS Windows firewall blocks ping by default.
- Your system is unplugged from the network.

Traceroute

traceroute command *traces the route* taken by the packets from your system to another host.

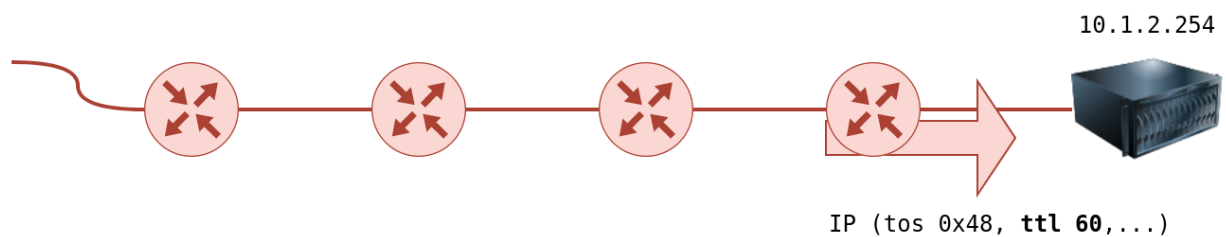
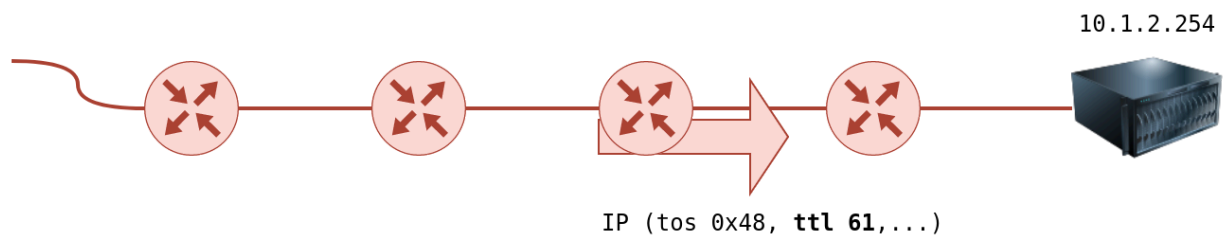
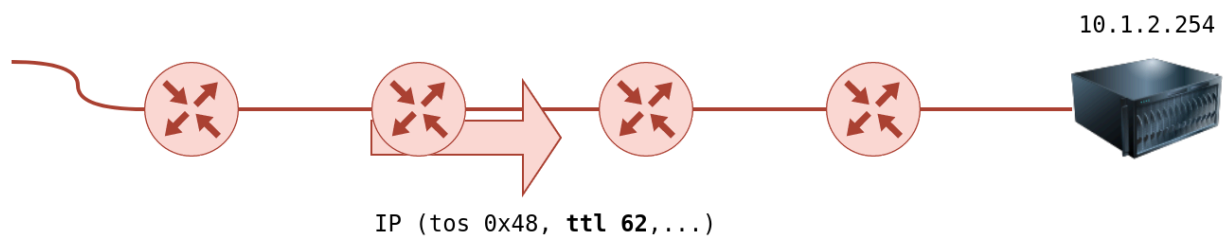
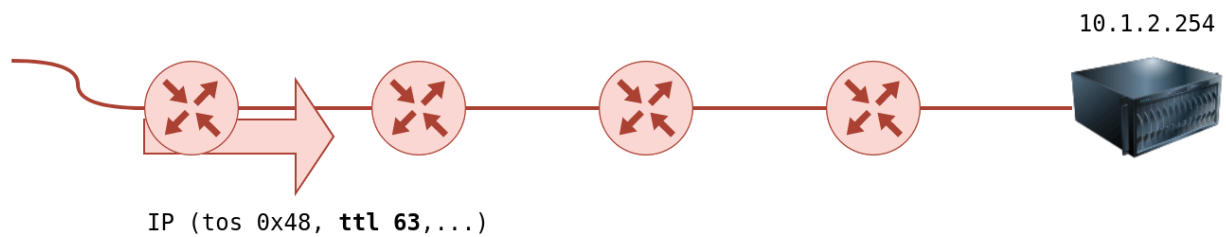
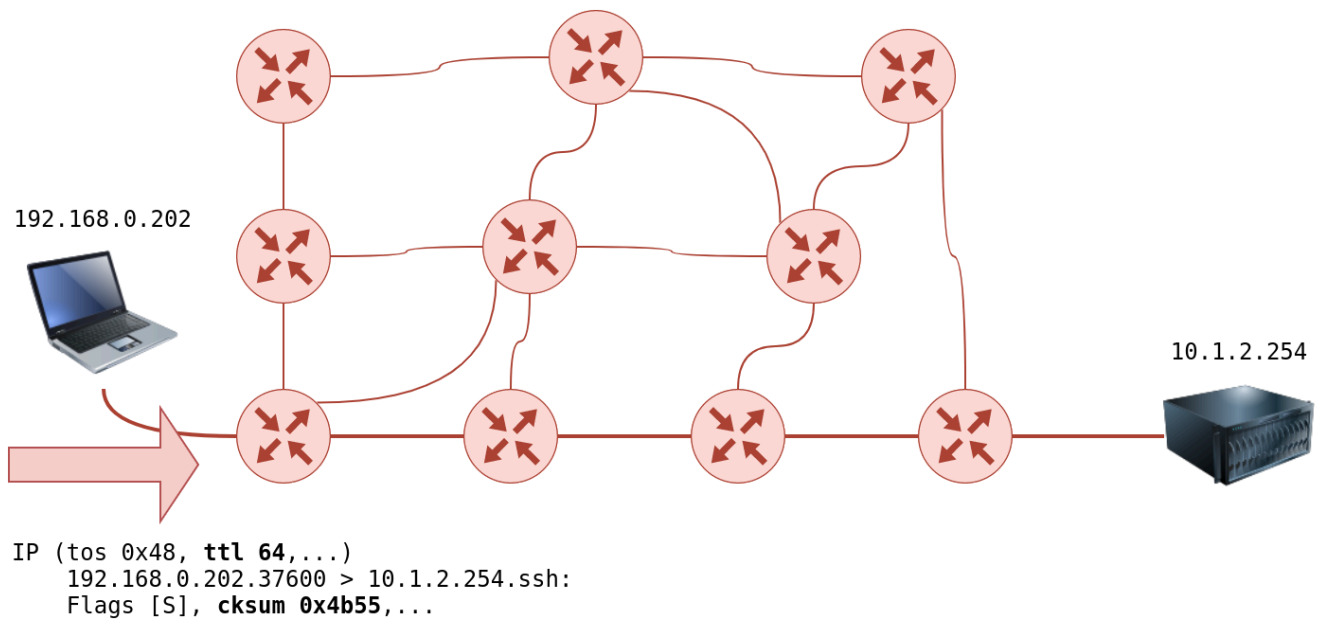
purpose of a traceroute is to find the IP addresses of the routers or hops that a packet traverses as it goes from your system to a target host. This command also reveals the number of routers between the two systems

On Linux and macOS, the command to use is

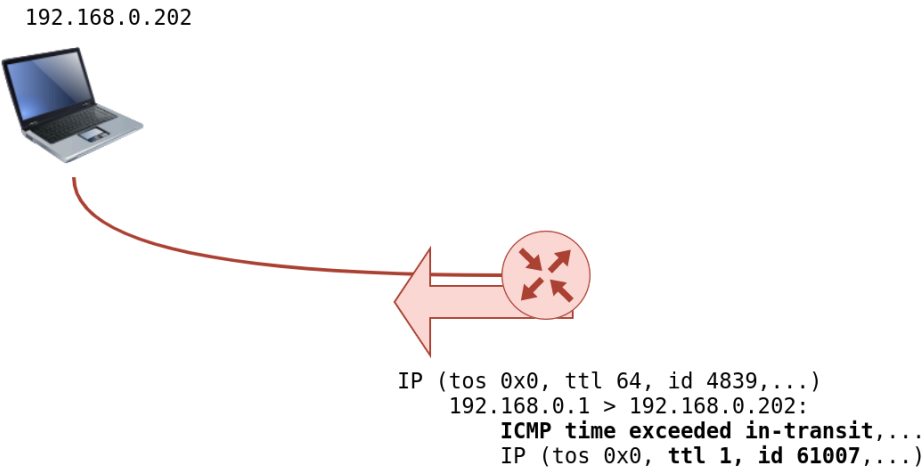
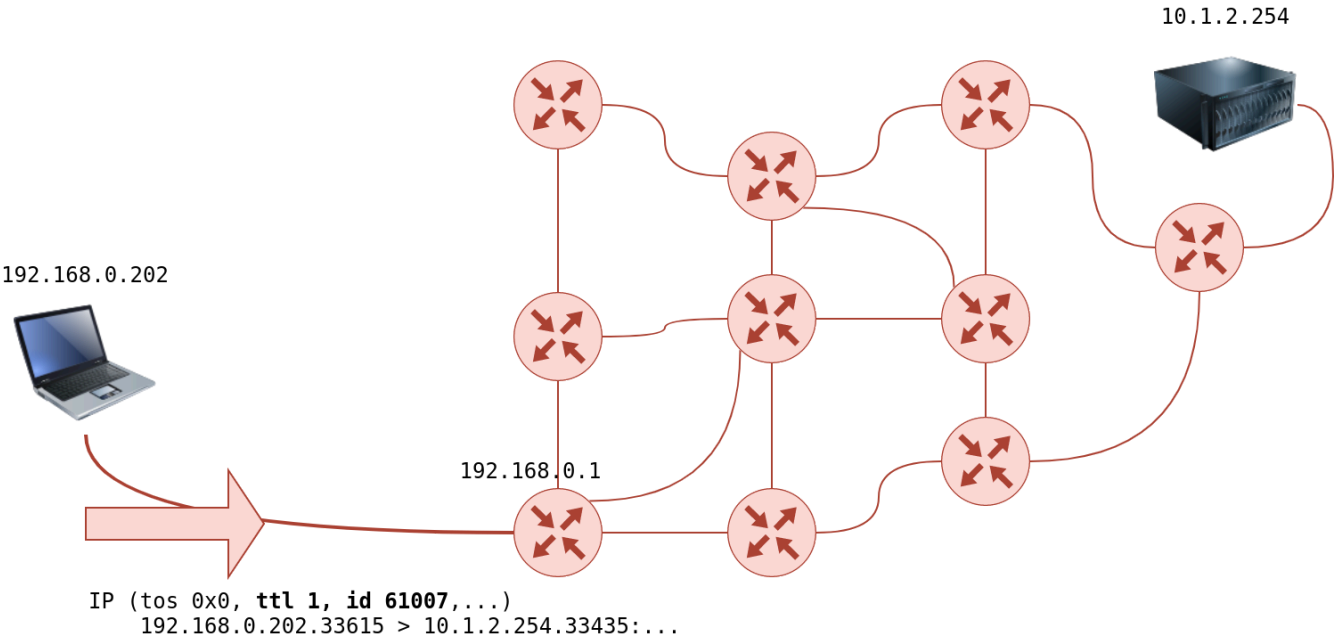
```
traceroute 10.129.183.187 ,
```

and on MS Windows, it is

```
tracert 10.129.183.187
```



if the TTL reaches 0, it will be dropped, and an ICMP Time-to-Live exceeded would be sent to the original sender.



```

user@AttackBox$ traceroute tryhackme.com
traceroute to tryhackme.com (172.67.69.208), 30 hops max, 60 byte packets
 1  ec2-3-248-240-5.eu-west-1.compute.amazonaws.com (3.248.240.5)  2.663 ms * ec2-3-
 2  100.66.8.86 (100.66.8.86)  43.231 ms 100.65.21.64 (100.65.21.64)  18.886 ms 100
 3  * 100.66.16.176 (100.66.16.176)  8.006 ms *
 4  100.66.11.34 (100.66.11.34)  17.401 ms 100.66.10.14 (100.66.10.14)  23.614 ms 1
 5  100.66.7.35 (100.66.7.35)  12.808 ms 100.66.6.109 (100.66.6.109)  14.791 ms *
 6  100.65.14.131 (100.65.14.131)  1.026 ms 100.66.5.189 (100.66.5.189)  19.246 ms
 7  100.65.13.143 (100.65.13.143)  14.254 ms 100.95.18.131 (100.95.18.131)  0.944 n
 8  100.95.2.143 (100.95.2.143)  0.680 ms 100.100.4.46 (100.100.4.46)  1.392 ms 100
 9  100.100.20.76 (100.100.20.76)  7.819 ms 100.92.11.36 (100.92.11.36)  18.669 ms
10  100.92.11.112 (100.92.11.112)  17.852 ms * 100.92.11.158 (100.92.11.158)  16.68
11  100.92.211.82 (100.92.211.82)  19.713 ms 100.92.0.126 (100.92.0.126)  18.603 ms
12  99.83.69.207 (99.83.69.207)  17.603 ms 15.827 ms 17.351 ms
13  100.92.9.83 (100.92.9.83)  17.894 ms 100.92.79.136 (100.92.79.136)  21.250 ms 1
14  172.67.69.208 (172.67.69.208)  17.976 ms 16.945 ms 100.92.9.3 (100.92.9.3)  17

```

In the traceroute output above, we have 14 numbered lines; each line represents one router/hop. Our system sends three packets with TTL set to 1, then three packets with TTL set to 2, and so forth. Depending on the network topology, we might get replies from up to 3 different routers, depending on the route taken by the packet. Consider line number 12, the twelfth router with the listed IP address has dropped the packet three times and sent an ICMP time exceeded in-transit message. The line `12 99.83.69.207 (99.83.69.207) 17.603 ms 15.827 ms 17.351 ms` shows the time in milliseconds for each reply to reach our system.

On the other hand, we can see that we received only a single reply on the third line. The two stars in the output `3 * 100.66.16.176 (100.66.16.176) 8.006 ms *` indicate that our system didn't receive two expected ICMP time exceeded in-transit messages.

Finally, in the first line of the output, we can see that the packets leaving the AttackBox take different routes. We can see two routers that responded to TTL being one. Our system never received the third expected ICMP message.

```

user@AttackBox$ traceroute tryhackme.com
traceroute to tryhackme.com (104.26.11.229), 30 hops max, 60 byte packets
 1  ec2-79-125-1-9.eu-west-1.compute.amazonaws.com (79.125.1.9)  1.475 ms * ec2-3-
 2  100.65.20.160 (100.65.20.160)  16.575 ms 100.66.8.226 (100.66.8.226)  23.241 m
 3  100.66.16.50 (100.66.16.50)  2.777 ms 100.66.11.34 (100.66.11.34)  22.288 ms 1
 4  100.66.6.47 (100.66.6.47)  17.264 ms 100.66.7.161 (100.66.7.161)  39.562 ms 10
 5  100.66.5.123 (100.66.5.123)  20.099 ms 100.66.7.239 (100.66.7.239)  19.253 ms
 6  * 100.66.5.223 (100.66.5.223)  16.172 ms 100.65.15.135 (100.65.15.135)  0.424
 7  100.65.12.135 (100.65.12.135)  0.390 ms 100.65.12.15 (100.65.12.15)  1.045 ms
 8  100.100.4.16 (100.100.4.16)  0.482 ms 100.100.20.122 (100.100.20.122)  0.795 m
 9  100.100.20.86 (100.100.20.86)  0.442 ms 100.100.4.78 (100.100.4.78)  0.347 ms
10  100.92.212.20 (100.92.212.20)  11.611 ms 100.92.11.54 (100.92.11.54)  12.675 m
11  100.92.6.52 (100.92.6.52)  11.427 ms 100.92.6.50 (100.92.6.50)  11.033 ms 100.
12  100.92.210.139 (100.92.210.139)  10.026 ms 100.92.6.13 (100.92.6.13)  14.586 m
13  100.92.79.12 (100.92.79.12)  12.011 ms 100.92.79.68 (100.92.79.68)  11.318 ms
14  100.92.9.27 (100.92.9.27)  11.354 ms 100.92.80.31 (100.92.80.31)  13.000 ms 52
15  150.222.241.85 (150.222.241.85)  9.660 ms 52.93.135.81 (52.93.135.81)  10.941
16  100.92.228.102 (100.92.228.102)  15.168 ms 100.92.227.41 (100.92.227.41)  10.1
17  100.92.232.111 (100.92.232.111)  10.589 ms 100.92.231.69 (100.92.231.69)  16.6
18  100.91.205.140 (100.91.205.140)  11.551 ms 100.91.201.62 (100.91.201.62)  10.2
19  100.91.205.79 (100.91.205.79)  11.112 ms 100.91.205.83 (100.91.205.83)  11.040
20  100.91.211.45 (100.91.211.45)  9.486 ms 100.91.211.79 (100.91.211.79)  13.693
21  100.100.6.81 (100.100.6.81)  11.522 ms 100.100.68.70 (100.100.68.70)  10.181 m
22  100.100.65.131 (100.100.65.131)  10.371 ms 100.100.92.6 (100.100.92.6)  10.939
23  100.100.2.74 (100.100.2.74)  15.317 ms 100.100.66.17 (100.100.66.17)  11.492 m
24  100.100.16.16 (100.100.16.16)  19.155 ms 100.100.16.28 (100.100.16.28)  19.147
25  99.83.89.19 (99.83.89.19)  28.929 ms * 21.790 ms
26  104.26.11.229 (104.26.11.229)  11.070 ms 11.058 ms 11.982 ms

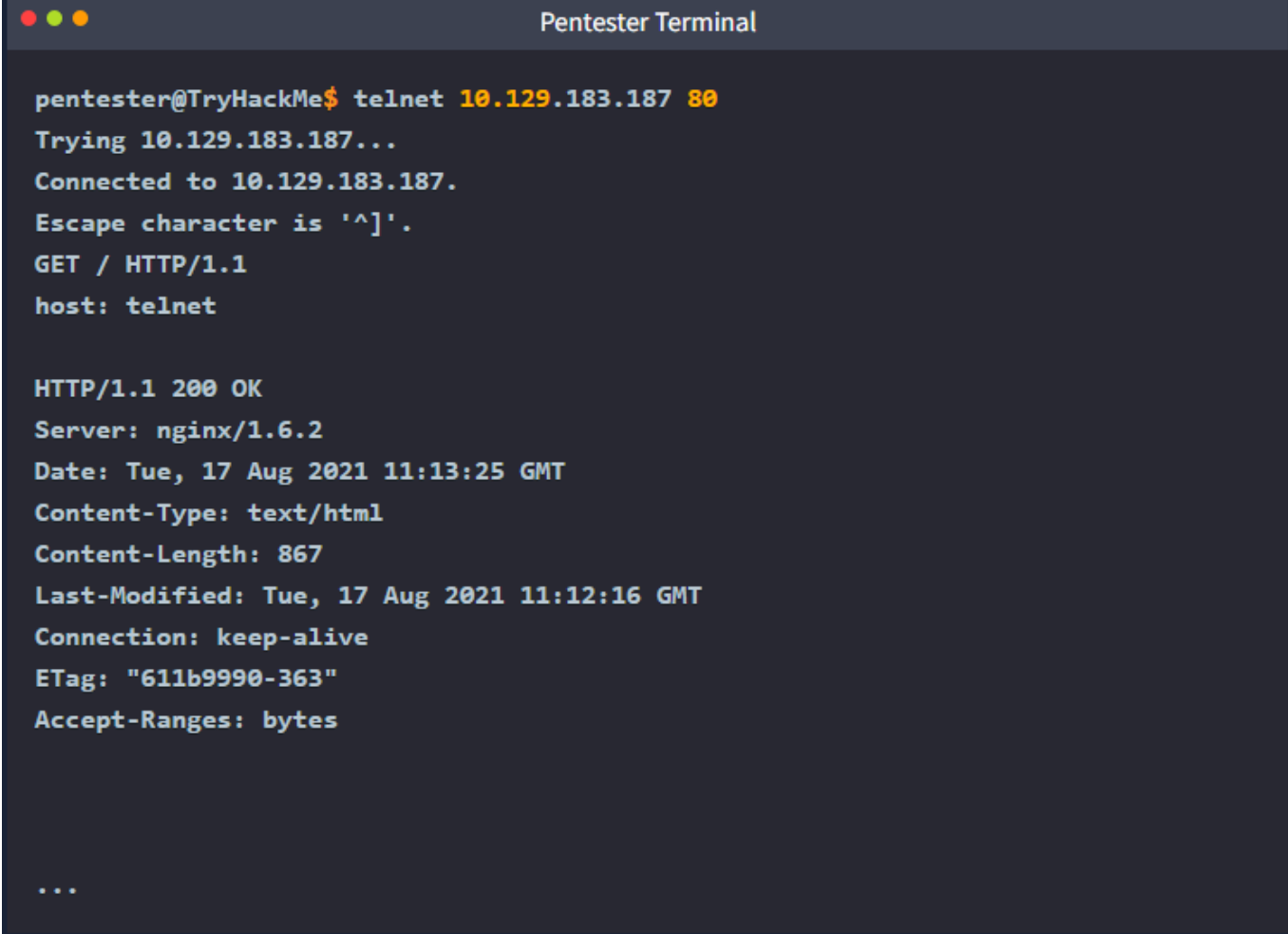
```

- The number of hops/routers between your system and the target system depends on the time you are running traceroute. There is no guarantee that your packets will always follow the same route, even if you are on the same network or you repeat the traceroute command within a short time.
- Some routers return a public IP address. You might examine a few of these routers based on the scope of the intended penetration testing.
- Some routers don't return a reply.

Telnet

The default port used by telnet is 23. From a security perspective, `telnet` sends all the data, including usernames and passwords, in cleartext. Sending in cleartext makes it easy for anyone, who has access to the communication channel, to steal the login credentials. The secure alternative is SSH (Secure Shell) protocol.

Knowing that telnet client relies on the TCP protocol, you can use Telnet to connect to any service and grab its banner. Using `telnet 10.129.183.187 PORT`, you can connect to any service running on TCP and even exchange a few messages unless it uses encryption.



```
Pentester Terminal

pentester@TryHackMe$ telnet 10.129.183.187 80
Trying 10.129.183.187...
Connected to 10.129.183.187.
Escape character is '^]'.
GET / HTTP/1.1
host: telnet

HTTP/1.1 200 OK
Server: nginx/1.6.2
Date: Tue, 17 Aug 2021 11:13:25 GMT
Content-Type: text/html
Content-Length: 867
Last-Modified: Tue, 17 Aug 2021 11:12:16 GMT
Connection: keep-alive
ETag: "611b9990-363"
Accept-Ranges: bytes

...
```

Of particular interest for us is discovering the type and version of the installed web server, `Server: nginx/1.6.2`. In this example, we communicated with a web server, so we used basic `HTTP` commands. If we connect to a mail server, we need to use proper commands based on the protocol, such as `SMTP` and `POP3`.

telnet on the target machine told me the name of the server and the version of the server


```

telnet: Unable to connect to remote host: Connection refused
root@ip-10-129-80-64:~# telnet 10.129.183.187 80
Trying 10.129.183.187...
Connected to 10.129.183.187.
Escape character is '^]'.
]
HTTP/1.1 400 Bad Request
Date: Wed, 15 Apr 2026 18:13:35 GMT
Server: Apache/2.4.61 (Debian)
Content-Length: 336
Connection: close
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>400 Bad Request</title>
</head><body>
<h1>Bad Request</h1>
<p>Your browser sent a request that this server could not understand.<br />
</p>
<hr>
<address>Apache/2.4.61 (Debian) Server at ip-10-129-183-187.eu-west-3.compute.internal Port 80</address>
</body></html>
Connection closed by foreign host.
root@ip-10-129-80-64:~#

```

Netcat

Netcat supports both TCP and UDP protocols. It can function as a client that connects to a listening port; alternatively, it can act as a server that listens on a port of your choice

```
nc 10.129.183.187 PORT
```

```

pentester@TryHackMe$ nc 10.129.183.187 80
GET / HTTP/1.1
host: netcat

HTTP/1.1 200 OK
Server: nginx/1.6.2
Date: Tue, 17 Aug 2021 11:39:49 GMT
Content-Type: text/html
Content-Length: 867
Last-Modified: Tue, 17 Aug 2021 11:12:16 GMT
Connection: keep-alive
ETag: "611b9990-363"
Accept-Ranges: bytes
...

```

Based on the output `Server: nginx/1.6.2` we received, we can tell that on port 80, we have Nginx version 1.6.2 listening for incoming connections.]

You can use netcat to listen on a TCP port and connect to a listening port on another system.

On the *server* system, where you want to open a port and listen on it, you can issue `nc -lp 1234` or better yet, `nc -vnlp 1234`, which is equivalent to `nc -v -l -n -p 1234`, as you

would remember from the [Linux](#) room. The exact order of the letters does not matter as long as the port number is preceded directly by `-p`.

option	meaning
-l	Listen mode
-p	Specify the Port number
-n	Numeric only; no resolution of hostnames via DNS
-v	Verbose output (optional, yet useful to discover any bugs)
-vv	Very Verbose (optional)
-k	Keep listening after client disconnects

Notes:

- the option `-p` should appear just before the port number you want to listen on.
- the option `-n` will avoid DNS lookups and warnings.
- port numbers less than 1024 require root privileges to listen on.

On the *client*-side, you would issue `nc 10.129.183.187 PORT_NUMBER`. Here is an example of using `nc` to echo. After you successfully establish a connection to the server, whatever you type on the client-side will be echoed on the server-side and vice versa.

Consider the following example. On the server-side, we will listen on port 1234. We can achieve this with the command `nc -vnlp 1234` (same as `nc -lvp 1234`). In our case, the listening server has the IP address `10.129.183.187`, so we can connect to it from the client-side by executing `nc 10.129.183.187 1234`. This setup would echo whatever you type on one side to the other side of the `TCP` tunnel. You can find a recording of the process below. Note that the listening server is on the left side of the screen.

Putting it all together

Command	Example
ping	<code>ping -c 10 10.129.183.187</code> on Linux or macOS
ping	<code>ping -n 10 10.129.183.187</code> on MS Windows
tracert	<code>tracert 10.129.183.187</code> on Linux or macOS
tracert	<code>tracert 10.129.183.187</code> on MS Windows
telnet	<code>telnet 10.129.183.187 PORT_NUMBER</code>
netcat as client	<code>nc 10.129.183.187 PORT_NUMBER</code>
netcat as server	<code>nc -lvp PORT_NUMBER</code>